

Sujith S

Applied Scientist | Agentic AI · Forecasting · Reinforcement Learning · Personalized GenAI · Deep Learning
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Summary

Published researcher and Computational Statistics student at VIT University (2022-2027) with applied experience building scalable agentic AI systems, forecasting pipelines, reinforcement learning agents, and personalized Generative AI products. Proficient in Python, PyTorch, TensorFlow, LangChain, and CrewAI; experienced designing modularized ML systems, large-scale data pipelines with Apache Spark, and production-grade forecasting and recommendation systems on AWS. Strong academic foundation in statistics, deep learning, and machine learning with a published peer-reviewed paper.

Education

Int.MSc Computational Statistics and Data Analytics

VIT University | 2022-2027

Coursework: Machine Learning, Deep Learning, Statistical Modeling, NLP, Data Structures & Algorithms, Distributed Systems, Database Systems

CS50 Programs

Harvard CS50 (Online)

Completed [CS50x](#), [CS50P](#), [CS50AI](#), [CS50SQL](#), and [CS50R](#) covering algorithms, AI, Python, and full-stack systems.

Skills

Languages: Python, Java, SQL, R, JavaScript, TypeScript

Machine Learning & AI: Agentic AI, Reinforcement Learning, Forecasting, Deep Learning, NLP, LLMs, Personalized GenAI, Recommendation Systems, Statistical Modeling, Model Optimization, Inference Optimization

Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LangGraph, CrewAI, AutoGen, Scikit-learn, XGBoost, LightGBM, Prophet, Statsmodels, FAISS

Data & Warehousing: Apache Spark, Apache Kafka, Delta Lake, Apache Airflow, PostgreSQL, Redis, MLflow, dbt

Cloud & Infrastructure: AWS (Lambda, S3, SageMaker, Redshift, EKS), Docker, CI/CD Pipelines, Linux/Unix

Research & Publications

Stock Market Analysis and Prediction Using Machine Learning

2025

[El Ruha International Conference on Scientific Research](#), pp. 730-740. Built a unified forecasting framework applying LSTM, GRU, Transformer, ARIMA, XGBoost, LightGBM, and Reinforcement Learning agents (DQN, PPO, A2C) for financial time-series prediction; designed feature engineering pipelines, evaluated 21 models for stock price forecasting, and distilled results into actionable trading recommendations. [\[GitHub\]](#)

Projects

Demand Forecasting System

[GitHub](#)

Python, Prophet, Statsmodels, Scikit-learn, XGBoost, FastAPI, PostgreSQL, Redis, MLflow, Docker, AWS

- Built a production-grade AI-powered demand forecasting platform applying time-series modeling (Prophet, ARIMA, SARIMA), ensemble learning (XGBoost, Scikit-learn), lag feature engineering, seasonal decomposition, and automated model retraining with drift detection; designed scalable REST inference APIs on AWS with staging-to-production deployment pipelines, MLflow experiment tracking, and real-time forecast dashboards for revenue and inventory prediction.

Multi-Agent Business Automation System

[GitHub](#)

Python, CrewAI, LangGraph, LangChain, AutoGen, FastAPI, PostgreSQL, Redis, Docker, AWS, Prometheus

- Architected a scalable agentic AI platform where specialized autonomous agents (Finance, Analytics, Operations, Support) collaborate under a Supervisor Agent for multi-step reasoning, RL-based policy optimization, and intelligent task routing; implemented contextual memory, workflow orchestration, KPI analytics, and sentiment analysis to automate complex business workflows at enterprise scale.

Recommendation Engine

[GitHub](#)

Python, PyTorch, TensorFlow, Scikit-learn, XGBoost, FAISS, Sentence Transformers, FastAPI, PostgreSQL, Redis, Docker

- Engineered a hybrid personalized recommendation system combining collaborative filtering, content-based filtering, and embedding-based vector similarity search via FAISS; built modularized model training pipelines in PyTorch and TensorFlow with online model updates, ensemble learning, and optimized low-latency inference APIs to generate personalized recommendations at scale.

Work Experience

Founder

[Shoot the Breeze](#) |

2023 - Present

- Owned end-to-end ML product development: designed agentic and forecasting features, wrote clear modularized Python code for model training and inference pipelines, and shipped production services on AWS with Docker, CI/CD automation, and monitoring for measurable customer impact.

App/Web Developer

[CropSky](#) | VIT-TBI Research & Innovation Hub |

2024 - Present

- Developed ML-powered backend systems and scalable REST APIs on cloud infrastructure at a university-incubated startup; built automated data pipelines querying large-scale structured data, and collaborated with engineers to ship ML model outputs into production applications with measurable seller impact.

Certifications & Achievements

- [Google Generative AI Leader](#)
- [Google Cloud Computing Foundations](#)
- [FreeCodeCamp Legacy Full Stack](#)
- [OCI 2025 Certified Developer Professional](#)
- AWS [Solutions Architect](#) and [Developer](#)
- [Software Engineer Certification - HackerRank](#)